



Qualification Pack

Remote Sensing Junior Analyst (Agriculture)

QP Code: CUTM/AGR/Q0501

Version: 1.0

NSQF Level: 5

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CUTM/AGR/Q0501: Remote Sensing Junior Analyst (Agriculture)

Brief Job Description

This advanced course transfers complex information into actionable insights. These analysts use their expertise in earth observation data to bring valuable insights to farmers and the agricultural industry as a whole which includes monitoring crop health and growth, soil and water management and weather forecasting that empowers farmers and ultimately contribute to a more sustainable and productive agricultural sector.

Personal Attributes

A Remote Sensing Junior Analyst (Agriculture) should be detail-oriented, analytical, and tech-savvy with a strong grasp of geospatial tools and agricultural practices. They must possess good observation skills, critical thinking, and data interpretation abilities. Curiosity, precision, and the ability to work independently or in teams are essential for success.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CUTM/AGR/N0501: Fundamentals of Remote Sensing](#)
2. [CUTM/AGR/N0502: Spatial analysis techniques and statistical methods for remote sensing data](#)
3. [CUTM/AGR/N0503: Applications of remote sensing in crop monitoring, and water management](#)
4. [CUTM/AGR/N0504: Understanding of crop physiology, growth stages, and modeling techniques with remote sensing data](#)
5. [CUTM/AGR/N0505: Introduction to machine learning algorithms for classification, regression, and prediction tasks in agriculture](#)
6. [CUTM/VSQ/N7802: Apply Health and Safety at Workplace](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Agriculture
Sub-Sector	Precision Farming
Occupation	Cartographers and Surveyors



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Country	India
NSQF Level	5
Credits	21
Aligned to NCO/ISCO/ISIC Code	NSQF
Minimum Educational Qualification & Experience	Completed 2nd year of UG (UG Diploma) (Biotechnology/ Bioinformatics/Agriculture/Geography/Environmental sciences/Botany/Life science) with 1 Year of experience OR 12th grade Pass (Biotechnology/ Bioinformatics/Agriculture/Geography/Environmental sciences/Biology/Life science) with 3 Years of experience OR Previous relevant Qualification of NSQF Level (4) with 3 Years of experience OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	21 Years
Last Reviewed On	NA
Next Review Date	22/10/2027
NSQF Approval Date	22/10/2024
Version	1.0
Reference code on NQR	QG-05-AG-03350-2024-V1-CUTM
NQR Version	1.0

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CUTM/AGR/N0501: Fundamentals of Remote Sensing

Description

This NOS provides foundational knowledge of remote sensing, covering the principles, types of sensors, data acquisition, and image interpretation. It equips learners with the skills to process and analyze satellite imagery for various applications such as agriculture, forestry, and urban planning, enabling informed decision-making through geospatial data.

Scope

The scope covers the following :

- This NOS covers the basic scope of understanding remote sensing principles, electromagnetic spectrum, sensor types, and data interpretation techniques. It enables learners to apply satellite imagery and geospatial data in fields like agriculture, environmental monitoring, disaster management, and urban development, laying the groundwork for advanced remote sensing analysis and applications

Elements and Performance Criteria

Comprehensively articulate principles, techniques and applications of remote sensing in agriculture

To be competent, the user/individual on the job must be able to:

PC1. Understanding of basic principles of remote sensing and its operation

PC2. • Introduction to electromagnetic spectrum, sensors, platforms and its interaction
• with the surface of Earth

Interpret remote sensing data in agricultural practices

To be competent, the user/individual on the job must be able to:

PC3. Digital Image acquisition and processing techniques to analyze and interpret remote sensing data using software.

PC4. Learn to address remote sensing technique in multiple areas of agricultural challenges which includes monitoring crop health and yield and managing pests and diseases

PC5. Determining soil status and nutrient content

PC6. Collecting and identifying accurate location data for field-based studies

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the basic principles and definitions of remote sensing.

KU2. the electromagnetic spectrum and its role in remote sensing.

KU3. different types of sensors and platforms (e.g., satellite, aerial, UAV).

KU4. the concepts of spatial, spectral, temporal, and radiometric resolution.

KU5. various remote sensing data types such as optical, thermal, and radar.

KU6. the process of data acquisition, pre-processing, and interpretation.



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- KU7.** the importance of ground truthing in remote sensing analysis.
- KU8.** the limitations and challenges in remote sensing such as cloud cover, sensor calibration, and data accessibility.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Communicate effectively with team members and supervisors about remote sensing tasks.
- GS2.** Read and interpret technical manuals, satellite data sheets, and documentation.
- GS3.** Use basic computer operations for data entry, documentation, and reporting.
- GS4.** Work independently or in a team to complete remote sensing assignments.
- GS5.** Manage time efficiently and adhere to project deadlines.
- GS6.** Analyze spatial data to identify patterns and extract useful information.
- GS7.** Apply remote sensing knowledge to solve real-world problems in areas like agriculture, forestry, and disaster management.
- GS8.** Stay updated with the latest technologies and trends in remote sensing.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Comprehensively articulate principles, techniques and applications of remote sensing in agriculture</i>	10	12	-	10
PC1. Understanding of basic principles of remote sensing and its operation	5	6	-	5
PC2. • Introduction to electromagnetic spectrum, sensors, platforms and its interaction • with the surface of Earth	5	6	-	5
<i>Interpret remote sensing data in agricultural practices</i>	20	28	-	20
PC3. Digital Image acquisition and processing techniques to analyze and interpret remote sensing data using software.	5	6	-	5
PC4. Learn to address remote sensing technique in multiple areas of agricultural challenges which includes monitoring crop health and yield and managing pests and diseases	5	6	-	5
PC5. Determining soil status and nutrient content	5	7	-	5
PC6. Collecting and identifying accurate location data for field-based studies	5	9	-	5
NOS Total	30	40	-	30



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0501
NOS Name	Fundamentals of Remote Sensing
Sector	Agriculture
Sub-Sector	
Occupation	Cartographers and Surveyors
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024

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CUTM/AGR/N0502: Spatial analysis techniques and statistical methods for remote sensing data

Description

This NOS focuses on applying spatial analysis techniques and statistical methods to interpret remote sensing data. It covers the use of GIS tools, statistical analysis, georeferencing, spatial interpolation, and classification accuracy assessment. The knowledge and skills developed enable effective extraction of meaningful insights from spatial data for various applications in environmental monitoring, urban planning, and agriculture.

Scope

The scope covers the following :

- This NOS covers the application of spatial and statistical techniques for analyzing remote sensing data. It includes working with GIS tools, spatial layers, georeferencing, interpolation, and accuracy assessment methods. The scope extends to practical use in land-use planning, environmental assessment, agriculture monitoring, and decision-making based on geospatial intelligence.

Elements and Performance Criteria

Understand core principles of smart agriculture

To be competent, the user/individual on the job must be able to:

- PC1.** Exploring the various technologies used in precision agriculture including remote sensing, drones etc.
- PC2.** Collecting data from soil, plant, weather and crop yield

Analyze geospatial data to determine and visualize the application of fertilizer, water irrigation and pesticides

To be competent, the user/individual on the job must be able to:

- PC3.** Analyzing and interpreting agricultural data and to make informed decisions
- PC4.** Utilizing data to determine variable rate applications of fertilizers, pesticides, and water irrigation in agricultural field.
- PC5.** Evaluating the economic and environmental benefits
- PC6.** Utilization of robots and drones for different activities in field like planting, weeding, and harvesting

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the basic concepts of spatial analysis and its role in remote sensing.
- KU2.** GIS tools and techniques for spatial data handling.
- KU3.** the importance of georeferencing and coordinate systems.
- KU4.** basic statistical methods used in image analysis, such as mean, median, standard deviation.



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- KU5.** spatial interpolation methods like IDW, Kriging, and their uses.
- KU6.** how to integrate attribute data with spatial features for deeper analysis.
- KU7.** the limitations and challenges in spatial and statistical analysis of remote sensing data.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Interpret technical manuals, statistical outputs, and spatial data reports.
- GS2.** Use digital tools and platforms to document, report, and present findings.
- GS3.** Demonstrate logical thinking and problem-solving in spatial data interpretation.
- GS4.** Operate GIS and remote sensing software for spatial and statistical analysis.
- GS5.** Handle spatial interpolation and trend detection using appropriate methods.
- GS6.** Maintain data integrity and confidentiality in analysis outputs.
- GS7.** Stay updated with current tools, trends, and techniques in spatial data science.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understand core principles of smart agriculture</i>	14	20	-	-
PC1. Exploring the various technologies used in precision agriculture including remote sensing, drones etc.	7	10	-	-
PC2. Collecting data from soil, plant, weather and crop yield	7	10	-	-
<i>Analyze geospatial data to determine and visualize the application of fertilizer, water irrigation and pesticides</i>	26	40	-	-
PC3. Analyzing and interpreting agricultural data and to make informed decisions	6	10	-	-
PC4. Utilizing data to determine variable rate applications of fertilizers, pesticides, and water irrigation in agricultural field.	6	10	-	-
PC5. Evaluating the economic and environmental benefits	9	10	-	-
PC6. Utilization of robots and drones for different activities in field like planting, weeding, and harvesting	5	10	-	-
NOS Total	40	60	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0502
NOS Name	Spatial analysis techniques and statistical methods for remote sensing data
Sector	Agriculture
Sub-Sector	
Occupation	Cartographers and Surveyors
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024

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CUTM/AGR/N0503: Applications of remote sensing in crop monitoring, and water management

Description

This NOS focuses on using remote sensing techniques for monitoring crop health, growth stages, and water resource management. It covers interpretation of vegetation indices, analysis of satellite imagery, and integration of spatial and field data. The skills developed help support precision agriculture, irrigation planning, and environmental sustainability through effective geospatial analysis.

Scope

The scope covers the following :

- This NOS encompasses the use of remote sensing data to assess crop conditions, detect stress, and plan irrigation. It includes analyzing satellite imagery, applying vegetation indices, and interpreting temporal patterns. The scope extends to improving agricultural productivity, managing water resources, and supporting data-driven decisions for sustainable farming and resource conservation.

Elements and Performance Criteria

Implement Real-Time Crop Health Monitoring for Precision Agriculture

To be competent, the user/individual on the job must be able to:

- PC1.** Understanding the basic principles of soil and water resources, their physical, chemical, and biological properties.
- PC2.** Gaining knowledge regarding the fundamentals and practices of sustainable soil and water management.
- PC3.** Ability to assess and monitor quality and implementing strategies for conservation of soil and water
- PC4.** Problem-solving skills for addressing soil and water resource management challenges.
- PC5.** Know about the importance of soil and water resources for food security, ecosystem health, and human well-being
- PC6.** Proper land use planning and watershed management

Analyze soil moisture for Optimization of Soil and Water Resource Management

To be competent, the user/individual on the job must be able to:

- PC7.** Evaluate appropriate techniques for land and water conservation to prevent soil erosion, improve fertility of soil and enhance quality of water
- PC8.** Know the environmental impact of agricultural practices and water resource management, and propose sustainable solutions to mitigate negative effects.
- PC9.** Effective interaction and recommendations to stakeholders, including farmers, policymakers, and community members, and engage in collaborative efforts for sustainable resource management.
- PC10.** Communication and collaboration skills in resource management
- PC11.** Problem-solving skills for addressing different challenges in soil and water resource management



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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the role of remote sensing in agriculture and water resource management.
- KU2.** how to identify crop health, stress, and growth stages using remote sensing tools.
- KU3.** how to use temporal data for monitoring crop cycles and water bodies.
- KU4.** the application of remote sensing in drought detection and early warning systems.
- KU5.** basic preprocessing and analysis techniques used in crop and water studies.
- KU6.** the importance of geospatial data in policy-making and agricultural decision support systems.
- KU7.** how remote sensing supports irrigation planning and water availability assessment.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and interpret technical documents, maps, and remote sensing reports.
- GS2.** Follow instructions for field data collection and software-based analysis.
- GS3.** Apply logical thinking in interpreting remote sensing data for practical use.
- GS4.** Manage time efficiently during monitoring activities and reporting tasks.
- GS5.** Identify crop stages and stress zones using multispectral and temporal data.
- GS6.** Integrate field data with remote sensing outputs for improved accuracy.
- GS7.** Stay updated with modern tools, platforms, and methods in agri-remote sensing.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Implement Real-Time Crop Health Monitoring for Precision Agriculture</i>	28	25	-	-
PC1. Understanding the basic principles of soil and water resources, their physical, chemical, and biological properties.	1	-	-	-
PC2. Gaining knowledge regarding the fundamentals and practices of sustainable soil and water management.	5	5	-	-
PC3. Ability to assess and monitor quality and implementing strategies for conservation of soil and water	5	5	-	-
PC4. Problem-solving skills for addressing soil and water resource management challenges.	4	5	-	-
PC5. Know about the importance of soil and water resources for food security, ecosystem health, and human well-being	4	5	-	-
PC6. Proper land use planning and watershed management	9	5	-	-
<i>Analyze soil moisture for Optimization of Soil and Water Resource Management</i>	12	35	-	-
PC7. Evaluate appropriate techniques for land and water conservation to prevent soil erosion, improve fertility of soil and enhance quality of water	2	4	-	-
PC8. Know the environmental impact of agricultural practices and water resource management, and propose sustainable solutions to mitigate negative effects.	2	9	-	-
PC9. Effective interaction and recommendations to stakeholders, including farmers, policymakers, and community members, and engage in collaborative efforts for sustainable resource management.	2	8	-	-
PC10. Communication and collaboration skills in resource management	1	9	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Problem-solving skills for addressing different challenges in soil and water resource management	5	5	-	-
NOS Total	40	60	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0503
NOS Name	Applications of remote sensing in crop monitoring, and water management
Sector	Agriculture
Sub-Sector	
Occupation	Cartographers and Surveyors
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024

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CUTM/AGR/N0504: Understanding of crop physiology, growth stages, and modeling techniques with remote sensing data

Description

This NOS focuses on understanding crop physiology and growth stages through remote sensing. It includes using satellite data and modeling tools to simulate crop development, monitor phenological stages, and assess crop health. The aim is to support precision agriculture and decision-making by integrating field observations with remote sensing-based crop modeling techniques

Scope

The scope covers the following :

- This NOS covers the application of remote sensing in studying crop physiology and identifying growth stages. It includes the use of crop simulation models and satellite data for predicting yield and monitoring health. The scope involves integrating temporal imagery with agronomic knowledge to enhance decision-making in agriculture, optimize resource use, and improve farm productivity.

Elements and Performance Criteria

Integrate Crop Physiology with Remote Sensing Analytics

To be competent, the user/individual on the job must be able to:

- PC1.** Understanding of agricultural systems, including like crop growth, soil status, weather, and different farm management practices.
- PC2.** Developing a strong foundation in the principles and techniques of machine learning and artificial intelligence, with an aim on applications in agriculture sector
- PC3.** Acquire skills in collecting and preprocessing diverse data from different sectors of agriculture, such as satellite imagery, sensor data, weather data, soil data and crop yield information

Analyze images and create predictive models for crop monitoring and management

To be competent, the user/individual on the job must be able to:

- PC4.** Image processing and sensor based crop monitoring and management
- PC5.** Predicting models for disease and pest management in agriculture
- PC6.** Automation and robotic systems for different agricultural practices such as planting, harvesting, and maintenance
- PC7.** Proper documentation for agricultural data
- PC8.** Implications of AI technologies to solve agricultural problems
- PC9.** Practical applications such as hands on projects and specific case studies
- PC10.** Ability to apply different learning algorithms
- PC11.** Knowledge of emerging technologies and understanding future challenges and opportunities in agricultural sector
- PC12.** Ability to communicate complex technical concepts to a range of stakeholders.
- PC13.** Collaboration skills for multidisciplinary team work



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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** basic crop physiology including photosynthesis, transpiration, and nutrient uptake.
- KU2.** spectral responses of crops at various growth stages across different bands.
- KU3.** crop modeling techniques using remote sensing and weather data.
- KU4.** environmental factors influencing crop growth (temperature, moisture, light).
- KU5.** the importance of remote sensing-based crop monitoring in improving decision-making in agriculture.
- KU6.** the different phenological stages of crop growth (e.g., germination, tillering, flowering, maturity).

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Interpret technical literature and manuals related to crop growth and remote sensing.
- GS2.** Follow instructions and standard procedures during crop modeling and data collection.
- GS3.** Apply critical thinking to interpret crop development trends and patterns.
- GS4.** Manage tasks efficiently within given timelines and quality standards.
- GS5.** Operate modeling tools such as DSSAT, WOFOST, or similar platforms for crop simulation.
- GS6.** Integrate environmental parameters and satellite data in crop models.
- GS7.** Follow ethical data practices and maintain confidentiality of sensitive agricultural data.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Integrate Crop Physiology with Remote Sensing Analytics</i>	8	15	-	-
PC1. Understanding of agricultural systems, including like crop growth, soil status, weather, and different farm management practices.	2	5	-	-
PC2. Developing a strong foundation in the principles and techniques of machine learning and artificial intelligence, with an aim on applications in agriculture sector	4	5	-	-
PC3. Acquire skills in collecting and preprocessing diverse data from different sectors of agriculture, such as satellite imagery, sensor data, weather data, soil data and crop yield information	2	5	-	-
<i>Analyze images and create predictive models for crop monitoring and management</i>	22	55	-	-
PC4. Image processing and sensor based crop monitoring and management	2	5	-	-
PC5. Predicting models for disease and pest management in agriculture	4	5	-	-
PC6. Automation and robotic systems for different agricultural practices such as planting, harvesting, and maintenance	1	5	-	-
PC7. Proper documentation for agricultural data	6	5	-	-
PC8. Implications of AI technologies to solve agricultural problems	1	5	-	-
PC9. Practical applications such as hands on projects and specific case studies	1	5	-	-
PC10. Ability to apply different learning algorithms	2	5	-	-
PC11. Knowledge of emerging technologies and understanding future challenges and opportunities in agricultural sector	2	5	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Ability to communicate complex technical concepts to a range of stakeholders.	2	5	-	-
PC13. Collaboration skills for multidisciplinary team work	1	10	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0504
NOS Name	Understanding of crop physiology, growth stages, and modeling techniques with remote sensing data
Sector	Agriculture
Sub-Sector	
Occupation	Cartographers and Surveyors
NSQF Level	5
Credits	2
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024

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CUTM/AGR/N0505: Introduction to machine learning algorithms for classification, regression, and prediction tasks in agriculture

Description

This NOS introduces fundamental machine learning techniques used in agriculture for classification, regression, and prediction tasks. It covers key algorithms, data preprocessing, and model evaluation methods. The focus is on applying ML tools to analyze agricultural data, support crop monitoring, yield prediction, and decision-making, thereby enhancing the efficiency and accuracy of precision farming practices.

Scope

The scope covers the following :

- This NOS focuses on applying machine learning techniques to solve agricultural challenges. It includes classification for crop type and disease detection, regression for yield and resource prediction, and forecasting for decision-making. The scope extends to handling diverse agricultural datasets, integrating remote sensing with ML, and using data-driven insights to enhance productivity and sustainability in modern farming.

Elements and Performance Criteria

Machine learning and its application in crop management

To be competent, the user/individual on the job must be able to:

- PC1.** ML algorithms usage for classification (e.g., Decision Trees, SVM, K-Nearest Neighbors), regression and prediction
- PC2.** Learn to gather and preprocess agricultural data, including handling missing values, data normalization.
- PC3.** Practice building models for crop classification using labeled data to identify crop types, soil types, or pest presence.
- PC4.** Apply regression techniques to predict crop yields based on variables like weather data, soil conditions, and crop growth stages
- PC5.** Create predictive models for irrigation, fertilization, and pest control scheduling. Evaluate model effectiveness
- PC6.** Implement cross-validation techniques to assess model robustness and use hyper parameter tuning to optimize model performance for agricultural applications.

Apply Machine learning for monitoring and crop yield prediction, pest management or optimal irrigation schedules

To be competent, the user/individual on the job must be able to:

- PC7.** ML algorithms to analyze drone images to detect crop health, pest infestations, and nutrient deficiencies
- PC8.** Models predict crop yields based on weather, soil, and historical data, aiding planning and supply chain management.
- PC9.** Optimization of water and nutrient application, minimizing waste and maximizing crop growth by analyzing soil and environmental conditions.

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PC10. ML-driven image recognition identifies and targets weeds and pests precisely, reducing pesticide use.

Manage projects and framework using machine learning

To be competent, the user/individual on the job must be able to:

- PC11.** Fundamental project management principles, methodologies, and frameworks
- PC12.** Ability to apply project management principles to various industry contexts
- PC13.** Demonstrating effective teamwork, conflict resolution, and leadership in project environments
- PC14.** Proficiency in demonstrating tools like Gantt charts, critical path analysis, and risk management
- PC15.** Identifying potential challenges associated with a project and develop strategies to mitigate them.
- PC16.** Evaluating the impact of risks on the outcomes and timelines of the project
- PC17.** Developing comprehensive plans for project, including tasks, timelines, and resource allocation
- PC18.** Understanding the importance of quality management in project submission
- PC19.** Implementing quality control measures to ensure the outcomes meet defined standards of the project
- PC20.** Understanding the ethical standards in project management and interacting with the team members, clients, and stakeholders

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the basic concepts and terminology of machine learning (ML).
- KU2.** different types of ML techniques supervised, unsupervised, and reinforcement learning.
- KU3.** classification algorithms (e.g., Decision Trees, Random Forest, SVM) for crop disease detection and land cover mapping.
- KU4.** regression models (e.g., Linear Regression, Ridge Regression) for predicting crop yield and environmental parameters.
- KU5.** the role of data preprocessing normalization, feature selection, and dimensionality reduction.
- KU6.** about the integration of remote sensing data with machine learning models in agriculture.
- KU7.** the ethical considerations and data privacy concerns in AI-based agricultural systems.
- KU8.** Stay updated with the trends in AI and ML applications for smart farming and precision agriculture.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Communicate effectively using technical terms related to machine learning and agriculture.
- GS2.** Work collaboratively in data analysis teams or interdisciplinary agricultural tech projects.
- GS3.** Present model results and findings clearly in reports or presentations.



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- GS4.** Adapt to evolving technologies in agri-tech and AI-based tools.
- GS5.** Preprocess datasets for model training, including cleaning, encoding, and normalization.
- GS6.** Evaluate model accuracy using appropriate statistical measures.
- GS7.** Maintain data privacy and follow ethical practices in model development.
- GS8.** Stay updated with emerging ML techniques relevant to precision agriculture.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Machine learning and its application in crop management</i>	9	13	-	10
PC1. ML algorithms usage for classification (e.g., Decision Trees, SVM, K-Nearest Neighbors), regression and prediction	1	2	-	2
PC2. Learn to gather and preprocess agricultural data, including handling missing values, data normalization.	2	-	-	2
PC3. Practice building models for crop classification using labeled data to identify crop types, soil types, or pest presence.	2	4	-	2
PC4. Apply regression techniques to predict crop yields based on variables like weather data, soil conditions, and crop growth stages	2	3	-	2
PC5. Create predictive models for irrigation, fertilization, and pest control scheduling. Evaluate model effectiveness	1	4	-	2
PC6. Implement cross-validation techniques to assess model robustness and use hyper parameter tuning to optimize model performance for agricultural applications.	1	-	-	-
<i>Apply Machine learning for monitoring and crop yield prediction, pest management or optimal irrigation schedules</i>	4	7	-	4
PC7. ML algorithms to analyze drone images to detect crop health, pest infestations, and nutrient deficiencies	1	3	-	2
PC8. Models predict crop yields based on weather, soil, and historical data, aiding planning and supply chain management.	2	2	-	2
PC9. Optimization of water and nutrient application, minimizing waste and maximizing crop growth by analyzing soil and environmental conditions.	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. ML-driven image recognition identifies and targets weeds and pests precisely, reducing pesticide use.	-	-	-	-
<i>Manage projects and framework using machine learning</i>	17	20	-	16
PC11. Fundamental project management principles, methodologies, and frameworks	2	2	-	2
PC12. Ability to apply project management principles to various industry contexts	1	3	-	2
PC13. Demonstrating effective teamwork, conflict resolution, and leadership in project environments	2	3	-	1
PC14. Proficiency in demonstrating tools like Gantt charts, critical path analysis, and risk management	2	3	-	1
PC15. Identifying potential challenges associated with a project and develop strategies to mitigate them.	1	3	-	1
PC16. Evaluating the impact of risks on the outcomes and timelines of the project	2	3	-	1
PC17. Developing comprehensive plans for project, including tasks, timelines, and resource allocation	2	3	-	2
PC18. Understanding the importance of quality management in project submission	2	-	-	2
PC19. Implementing quality control measures to ensure the outcomes meet defined standards of the project	2	-	-	2
PC20. Understanding the ethical standards in project management and interacting with the team members, clients, and stakeholders	1	-	-	2
NOS Total	30	40	-	30



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National Occupational Standards (NOS) Parameters

NOS Code	CUTM/AGR/N0505
NOS Name	Introduction to machine learning algorithms for classification, regression, and prediction tasks in agriculture
Sector	Agriculture
Sub-Sector	
Occupation	Cartographers and Surveyors
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024

Qualification Pack

CUTM/VSQ/N7802: Apply Health and Safety at Workplace

Description

This NOS pertains to ensuring a safe, hazard-free work environment by performing safety checks, using PPE, understanding chemical hazards, and responding to emergencies. It also includes handling first aid, waste disposal, risk assessment, and documenting safety incidents and market data as per workplace protocols.

Scope

The scope covers the following :

- The scope of this NOS includes practicing safety measures in day-to-day operations, preventing accidents, using and maintaining emergency and safety equipment, and following emergency response procedures. It also covers first aid administration, proper documentation, and handling hazardous materials responsibly to maintain workplace health and safety standards.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Undertaking basic safety checks before operation of all machinery and vehicles and hazards are reported to the appropriate supervisor.
- PC2.** Work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy.
- PC3.** Reading and understanding the hazards of use and contamination mentioned on the labels of pesticides/fumigants etc
- PC4.** Assessing the risks prior to performing manual handling jobs and work according to currently recommended safe practice.
- PC5.** Use equipment and materials safely and correctly and return the same to designated storage when not in use.
- PC6.** Dispose of waste safely and correctly in a designated area.
- PC7.** Recognize risks to bystanders and take action to reduce risk associated with jobs in the workplace.
- PC8.** Perform your work in a manner which minimizes environmental damage all procedures and work instructions for controlling risk are followed closely.
- PC9.** Report any accidents, incidents or problems without delay to an appropriate person and take necessary immediate action to reduce further danger.
- PC10.** Follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.
- PC11.** Follow emergency procedures to company standard / workplace requirements.
- PC12.** Ascertain availability and non-availability of specific market information.
- PC13.** Perform documentation for analyzing market information.
- PC14.** Evaluate the authenticity of information received.

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- PC15.** Analyze the information for taking decision.
- PC16.** Use emergency equipment in accordance with manufacturers' specifications and workplace requirements.
- PC17.** Provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.
- PC18.** Recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate.
- PC19.** Details report of first aid administered in accordance with workplace procedures.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** basic safety protocols for machinery and vehicle operation in agricultural or industrial settings.
- KU2.** the types of personal protective equipment (PPE) required for specific tasks and their correct usage.
- KU3.** hazard symbols and labels, particularly on chemicals like pesticides or fumigants.
- KU4.** the manuals handling risks and apply appropriate ergonomic principles to reduce injury.
- KU5.** the procedures for safe use and storage of tools, equipment, and materials.
- KU6.** waste disposal protocols as per environmental and workplace guidelines.
- KU7.** potential hazards to coworkers and bystanders, and know mitigation strategies.
- KU8.** emergency procedures, including fire safety, accident response, and first aid.
- KU9.** documentation processes for safety and market information.
- KU10.** evaluate, verify, and use market and safety information for decision-making.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Perform safety checks independently and communicate hazards effectively to supervisors.
- GS2.** Use and maintain equipment responsibly, ensuring safety and longevity.
- GS3.** Demonstrate the ability to follow written and verbal instructions related to hazardous materials.
- GS4.** Apply decision-making skills in risk assessment and emergency scenarios.
- GS5.** Communicate clearly during emergencies, including providing accurate directions.
- GS6.** Execute emergency treatment using recognized first aid techniques.
- GS7.** Maintain accurate and timely documentation for safety incidents and market-related data.
- GS8.** Handle emergency equipment confidently, adhering to operational protocols.
- GS9.** Evaluate and interpret workplace data for taking informed decisions.
- GS10.** Exhibit responsibility and environmental awareness in performing daily tasks.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	40	-	30
PC1. Undertaking basic safety checks before operation of all machinery and vehicles and hazards are reported to the appropriate supervisor.	2	3	-	2
PC2. Work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy.	2	3	-	2
PC3. Reading and understanding the hazards of use and contamination mentioned on the labels of pesticides/fumigants etc	2	3	-	2
PC4. Assessing the risks prior to performing manual handling jobs and work according to currently recommended safe practice.	2	3	-	2
PC5. Use equipment and materials safely and correctly and return the same to designated storage when not in use.	2	3	-	2
PC6. Dispose of waste safely and correctly in a designated area.	-	3	-	-
PC7. Recognize risks to bystanders and take action to reduce risk associated with jobs in the workplace.	2	3	-	2
PC8. Perform your work in a manner which minimizes environmental damage all procedures and work instructions for controlling risk are followed closely.	2	3	-	2
PC9. Report any accidents, incidents or problems without delay to an appropriate person and take necessary immediate action to reduce further danger.	2	3	-	2
PC10. Follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.	2	3	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Follow emergency procedures to company standard / workplace requirements.	2	1	-	2
PC12. Ascertain availability and non-availability of specific market information.	2	1	-	2
PC13. Perform documentation for analyzing market information.	2	1	-	2
PC14. Evaluate the authenticity of information received.	-	1	-	-
PC15. Analyze the information for taking decision.	2	1	-	2
PC16. Use emergency equipment in accordance with manufacturers' specifications and workplace requirements.	2	1	-	2
PC17. Provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.	2	1	-	2
PC18. Recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate.	-	1	-	-
PC19. Details report of first aid administered in accordance with workplace procedures.	-	2	-	-
NOS Total	30	40	-	30



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CUTM/VSQ/N7802
NOS Name	Apply Health and Safety at Workplace
Sector	Cross Sectoral
Sub-Sector	
Occupation	Health and Safety
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Written Test

o Format: (i) True / False Statements and (ii) Multiple Choice Questions (MCQs)

o Objective: To assess the candidate's theoretical understanding of remote sensing principles, geospatial techniques, crop modeling, machine learning, and workplace safety.

2. Practical Test

o Format: Test tasks or project work based on real-world or simulated case studies.

o Objective: To evaluate the candidate's ability to apply remote sensing techniques, perform spatial and statistical analyses, operate GIS/RS tools, and use ML models in an agricultural context. Observation of problem-solving skills, software handling, and task execution will be emphasized.

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3. Structured Interview (Viva)

o Format: Oral interview or Q&A with the assessor.

o Objective: To assess conceptual clarity, technical justification, decision-making skills, and understanding of data-driven agricultural practices and workplace behavior.

Theory components will cover:

- Principles and types of remote sensing (optical, thermal, radar, etc.)
- Fundamentals of electromagnetic spectrum, satellite sensors, and image acquisition
- Statistical and spatial analysis methods including classification, and interpolation
- Agricultural applications such as crop health monitoring, irrigation planning, and yield estimation
- Crop physiology, phenology, and growth stage indicators
- Basics of machine learning algorithms like decision trees and regression for agri-data
- Safety measures in fieldwork and lab-based analysis
- Foundational employability skills including communication, teamwork, digital literacy, and time management

Practical components will cover:

- Process satellite imagery using GIS/remote sensing software
- Perform vegetation index calculations to assess crop health
- Apply statistical tools to interpret remote sensing outputs for agricultural insights
- Demonstrate application of machine learning models for crop classification or yield prediction
- Follow field safety protocols and demonstrate hazard awareness in data collection settings
- Perform tasks involving communication and data presentation in a team environment

Viva session will cover:

- Explain remote sensing outputs and justify interpretation decisions
- Describe real-world applications of techniques used during practicals



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- Discuss selection and interpretation of machine learning outputs
- Reflect on teamwork, communication, and safety awareness during assigned tasks

Minimum Passing Criteria:

1. 70% overall score is required to pass the qualification.
2. Minimum 60% score is required in each NOS (inclusive of theory, practical, and viva).

Reassessment Policy:

1. Candidates failing the assessment will be given one reassessment opportunity.
2. Reassessment will be conducted only after undergoing additional remedial training on the failed NOS components.

Assessor Details

1. Must be ToA (Training of Assessors) Certified.
2. Should be trained in assessment methodology, ethics, and NSQF-aligned evaluation practices

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 60

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CUTM/AGR/N0501.Fundamentals of Remote Sensing	30	40	-	30	100	10
CUTM/AGR/N0502.Spatial analysis techniques and statistical methods for remote sensing data	40	60	-	-	100	20
CUTM/AGR/N0503.Applications of remote sensing in crop monitoring, and water management	40	60	-	-	100	20
CUTM/AGR/N0504.Understanding of crop physiology, growth stages, and modeling techniques with remote sensing data	30	70	-	-	100	10
CUTM/AGR/N0505.Introduction to machine learning algorithms for classification, regression, and prediction tasks in agriculture	30	40	-	30	100	20
CUTM/VSQ/N7802.Apply Health and Safety at Workplace	30	40	-	30	100	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	220	340	-	90	650	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.